

Theta, Vector, and Unified-Action Constraints in TRM/TQM V3.0

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Abstract

This paper isolates the TRM/TQM V3.0 theta-observable path, vector frame-dragging candidate sector, and unified effective-action roadmap [3, 5, 4, 2]. The objective is to state what is currently tested-effective, what is structurally constrained, and what remains open at theorem level. The chain $\Theta \rightarrow O_5 \rightarrow \lambda_\Theta \rightarrow g_{\text{obs}}$ is strongly supported by guard tests; the vector sector is strongly hardened in weak-field candidate form through FD01-FD20; and unified-action guards UF01-UF09 preserve sector limits and bounded cross-couplings. These results remain below theorem-level first-principles closure and do not claim GR replacement.

1 Introduction

TRM/TQM V3.0 uses sector-wise hardening and cross-sector consistency guards to constrain model freedom while keeping claim boundaries explicit [2]. This paper focuses on the three tracks that define current cross-sector structure:

1. the nonlocal theta-observable chain,
2. the vector rotational/frame-dragging candidate sector,
3. the unified effective-action roadmap that links scalar, vector, and theta sectors.

2 Theta-Observable Path

2.1 Core chain

$$\Theta \rightarrow O_5 \rightarrow \lambda_\Theta \rightarrow g_{\text{obs}}.$$

2.2 Current test support

The theta path is currently hardened by TO01-TO28, TQK01-TQK04, LC01-LC08, and TOL01-TOL04 [3, 2]. At this stage, the chain is treated as tested-effective with strong guard-level support for operator behavior, holdout discipline, anti-proxy safeguards, and lattice-energy consistency.

2.3 Claim boundary

- Supported: strongly constrained theorem path at tested-effective level.
- Not claimed: theorem-level microscopic closure of the full $\Theta \rightarrow g_{\text{obs}}$ map.

3 Vector Frame-Dragging Candidate Sector

3.1 Core structure

The vector sector is based on:

$$\vec{A}_T, \quad \vec{B}_T = \nabla \times \vec{A}_T,$$

with weak-field frame-dragging candidate interpretation [5, 1].

3.2 Current hardening status

FD01-FD20 currently support [5, 2]:

1. rotational source generation and spin-sign reversal behavior,
2. weak-field scaling shape and null/spin-zero fallback consistency,
3. derived non-fitted effective k_T normalization discipline,
4. ablation/holdout/unit-scaling robustness,
5. controlled systematic-bias auditing under no-refit constraints.

3.3 Claim boundary

- Supported: tested-effective weak-field frame-dragging candidate sector.
- Not claimed: theorem-level first-principles closure or full quantitative GR-equivalence.

4 Unified Effective-Action Roadmap

4.1 Canonical roadmap form

$$S_{\text{eff}}[T, \vec{A}_T, \Theta] = S_T[T] + S_A[\vec{A}_T] + S_\Theta[\Theta] + S_{\text{int}}[T, \vec{A}_T, \Theta].$$

4.2 Current guard status

UF01-UF09 currently support [4, 2]:

1. scalar/vector/theta limit recovery,
2. zero-coupling additive decomposition,
3. bounded small cross-couplings,
4. global cross-coupling identifiability without per-group refit dependence,
5. preservation of MC/FD/TO guard behavior under small couplings.

4.3 Claim boundary

- Supported: candidate-level unified effective-action roadmap with guard preservation.
- Not claimed: theorem-level unification.

5 Cross-Sector Falsification Criteria

The current theta/vector/unified interpretation should be narrowed or rejected if:

1. theta-chain anti-proxy or holdout guarantees collapse under independent tests;
2. vector weak-field guard behavior fails under robust ablation/benchmark stress;
3. unified-action interaction terms break sector-limit recovery or MC/FD/TO guard preservation;
4. claimed cross-coupling structure requires hidden per-dataset re-normalization.

6 Remaining First-Principles Gaps

Open items remain explicit [3, 4, 5]:

- theorem-level microscopic closure for the theta chain,
- theorem-level first-principles closure for vector-sector normalization and benchmark mapping,
- theorem-level closure of unified cross-sector interaction structure.

7 Conclusion

In V3.0, theta, vector, and unified-action blocks are strongly constrained at tested-effective level and form a coherent candidate multi-sector framework under explicit guard discipline. The status is review-ready and falsifiable, but it is not theorem-level complete and does not claim replacement of General Relativity.

References

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